

BREA NAKATA

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Available for Co-op: January – June 2027

EDUCATION

Northeastern University

Khoury College of Computer Science

Boston, MA

Expected May 2029

Candidate for Bachelor of Science, Major: Computer Science, GPA: 3.65/4.0

Related Courses: Discrete Structures, Program Design & Implementation 1 & 2 (OOD), Computer Systems

Awards: NCWIT Aspirations in Computing National Honorable Mention; AFCEA Hawai'i STEM Scholarship

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript

Machine Learning & Computer Vision: PyTorch, TensorFlow, YOLO, Ultralytics, OpenCV

Tools & Platforms: Git, Linux, Docker, Flask, Selenium, REST APIs, NVIDIA Jetson, ArduPilot

EXPERIENCE

ML/Computer Vision Software Engineering Intern, DARPA Triage Challenge Team – Remote *Jan 2025 – Present*

- Owned the perception pipeline converting camera detections into real-world GPS coordinates on a 5-person team building an autonomous search-and-rescue drone for DARPA's DTC Phase 3 Finals
- Trained and fine-tuned YOLO-based object-detection models in PyTorch on a dataset of 1,500+ annotated images, reaching 77% mAP@50 on validation data for casualties, deployed on NVIDIA Jetson for real-time inference
- Built sensor and signal-processing code that estimates vital signs, fusing multi-source sensor data with injury classifications into a single triage assessment per casualty for submission
- Integrated the system with competition scoring API and a web-based operator interface, and wrote Python tooling to parse and analyze flight logs, diagnosing failures across 40+ test flights

Front-End Software Engineering Intern, University of Hawai'i: Project Hokulani – Honolulu, HI *Jun 2024 – Jul 2024*

- Built the front end for Parrot Forms, a 4-stage AI pipeline that transcribes doctor-patient conversations with Whisper, structures them into JSON with an LLM, and auto-fills patient intake forms
- Built a review step into the interface so clinicians confirm the LLM's extracted fields before submission, preventing silent extraction errors from reaching a patient's record; contributed Selenium automation populating 15+ intake fields from generated JSON

PROJECTS

Evaluating an AI-Generated Image Detector | PyTorch, torchvision, Hugging Face

Sep 2026 – Present

- Building a real-vs-AI image detector (ResNet18) on a 70K-image paired dataset spanning 6 text-to-image generators, focused on testing how far its accuracy can be trusted rather than maximizing it
- Designing experiments to measure generalization to generators unseen during training and robustness to JPEG compression, resizing, and blur

CookYourBooks – Recipe Management Application | Java, JavaFX, JUnit

Apr 2026

- Owned the Library View for a 3-person team-developed desktop application, implementing collection browsing, filtering, creation/deletion, navigation, and undo functionality using JavaFX and a ViewModel architecture
- Collaborated on additional features including multi-page recipe importing and a step-by-step cooking interface, integrating functionality across a shared Java codebase, delivering a fully functional application

ACTIVITIES & INTERESTS

Media Chair (prev. Freshman Representative), Northeastern Hawai'i Club – Boston, MA

Nov 2025 – Present

- Designed promotional graphics and managed club social media; choreographed a hula performance with 15+ members for the club's main annual cultural event

Interests: Japanese (bilingual) • Digital and traditional visual art (Scholastic Art & Writing Gold Key; American Visions Award nominee)